

SolutionMidtermExam

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Autumn 2025

1. Let us consider the vectors

$$a = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \quad c = \begin{bmatrix} -1 \\ -1 \\ 3 \end{bmatrix}, \quad d = \begin{bmatrix} 0 \\ 4 \\ 6 \end{bmatrix},$$

and respond to the following questions by providing enough reasoning:

- (a) (1 point) Compute $\|a\|^2 + \|c + d\| + \langle a, b \rangle$ and $\langle a + b, c - d \rangle$.
- (b) (1 point) Are the vectors a and b independent? Are they generating $\mathbb{R}^3$?
- (c) (1 point) Are the vectors a, b and c independent? Are they generating $\mathbb{R}^3$?
- (d) (2 point) Are the vectors a, b and d generating a basis for $\mathbb{R}^3$?

Answer.

(a) We compute:

$$\|a\|^2 = 2^2 + 1^2 + 0^2 = 5, \quad c + d = \begin{bmatrix} -1 \\ 3 \\ 9 \end{bmatrix}, \quad \|c + d\| = \sqrt{(-1)^2 + 3^2 + 9^2} = \sqrt{91}.$$

Also,

$$\langle a, b \rangle = 2 \cdot 1 + 1 \cdot 0 + 0 \cdot 3 = 2.$$

Thus

$$\|a\|^2 + \|c + d\| + \langle a, b \rangle = 5 + \sqrt{91} + 2 = 7 + \sqrt{91}.$$

Next,

$$a + b = \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}, \quad c - d = \begin{bmatrix} -1 \\ -5 \\ -3 \end{bmatrix},$$

so,

$$\langle a + b, c - d \rangle = 3(-1) + 1(-5) + 3(-3) = -3 - 5 - 9 = -17.$$

(b) To study independence of a and b , solve

$$\alpha a + \beta b = 0.$$

This gives the system

$$\begin{cases} 2\alpha + \beta = 0, \\ \alpha = 0, \\ 3\beta = 0, \end{cases}$$

which implies $\alpha = \beta = 0$. Therefore, a and b are *linearly independent*.

Since two independent vectors in $\mathbb{R}^3$ cannot span $\mathbb{R}^3$, they do not generate $\mathbb{R}^3$.

(c) Consider the combination

$$a - b + c = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

So, a, b, c are *linearly dependent*. Thus, they do *not* form a basis and do not generate $\mathbb{R}^3$.

(d) Consider the matrix whose columns are a, b, d :

$$M = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 4 \\ 0 & 3 & 6 \end{bmatrix}.$$

Compute the determinant:

$$\det(M) = 2 \det \begin{bmatrix} 0 & 4 \\ 3 & 6 \end{bmatrix} - 1 \det \begin{bmatrix} 1 & 4 \\ 0 & 6 \end{bmatrix} = 2(0 \cdot 6 - 4 \cdot 3) - (1 \cdot 6 - 4 \cdot 0) = 2(-12) - 6 = -24 - 6 = -30 \neq 0.$$

Since the determinant is nonzero, the vectors are linearly independent and form a basis of $\mathbb{R}^3$.

Thus a, b, d generate a basis for $\mathbb{R}^3$.

2. Let us consider the matrix

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 & 9 \\ 0 & -1 & 2 \\ 0 & 0 & -5 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 4 & 7 & 0 & 1 \\ 1 & 4 & 0 & 0 \\ 2 & 3 & 0 & 5 \end{bmatrix}, \quad D = \begin{bmatrix} 10 & 3 & -6 & 1 \\ 0 & 7 & 2 & 3 \\ 1 & 2 & 5 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

- (a) (2 points) Are the matrices A, B, C and D invertible? Provide reasoning.
 (b) (2 points) If A is invertible, calculate its inverse A^{-1} by Gauss-Jordan elimination and by Cramer's rule.
 (c) (1 point) How many solutions has the system $Ax = b$? (Provide reasoning) If it has a solution, compute it.

Answer.

- (a) To check matrix A , compute its determinant:

$$\det(A) = 1 \cdot \begin{vmatrix} 3 & 0 \\ 2 & 1 \end{vmatrix} = 1(3 \cdot 1 - 2 \cdot 0) = 3 \neq 0.$$

Thus, A is invertible.

Matrix B is upper triangular. Its determinant is

$$\det(B) = 1 \cdot (-1) \cdot (-5) = 5 \neq 0.$$

Thus, B is invertible.

Matrix C has a column of zeros. Therefore, $\det(C) = 0$, so, C is not invertible.

Matrix D :

$$\begin{aligned} \det(D) &= 2 \cdot (-1)^{4+4} \begin{vmatrix} 10 & 3 & -6 \\ 0 & 7 & 2 \\ 1 & 2 & 5 \end{vmatrix} = 2 \cdot \begin{vmatrix} 10 & 3 & -6 \\ 0 & 7 & 2 \\ 1 & 2 & 5 \end{vmatrix} = 2 \left(10 \begin{vmatrix} 7 & 2 \\ 2 & 5 \end{vmatrix} + 1 \begin{vmatrix} 3 & -6 \\ 7 & 2 \end{vmatrix} \right) \\ &= 2 \left(10(35 - 4) + 1(6 + 42) \right) = 2(310 + 48) = 2 \times 358 = 716. \end{aligned}$$

Thus, matrix D is invertible.

- (b) **Computing the inverse of A .**

$$\begin{aligned} [A|I] &= \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow[r_2 \rightarrow \frac{1}{2}r_2]{r_3 \rightarrow r_3 - r_1} \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & -3 & 0 & -1 & 0 & 1 \end{array} \right] \\ &\xrightarrow[r_3 \rightarrow r_3 + 3r_2]{r_1 \rightarrow r_1 - 3r_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{3}{2} & 1 & -\frac{3}{2} & 0 \\ 0 & 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{3}{2} & -1 & \frac{3}{2} & 1 \end{array} \right] \xrightarrow{r_3 \rightarrow \frac{2}{3}r_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{3}{2} & 1 & -\frac{3}{2} & 0 \\ 0 & 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{2}{3} & 1 & \frac{2}{3} \end{array} \right] \\ &\xrightarrow[r_2 \rightarrow r_2 - \frac{1}{2}r_3]{r_1 \rightarrow r_1 + \frac{3}{2}r_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & \frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & 0 & 1 & -\frac{2}{3} & 1 & \frac{2}{3} \end{array} \right] \end{aligned}$$

Thus,

$$A^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{3} & 0 & -\frac{1}{3} \\ -\frac{2}{3} & 1 & \frac{2}{3} \end{bmatrix}.$$

Cramer's rule computation:

Finding A^{-1} using Cofactors and Adjugate:

Step 1: Compute the determinant of A :

$$\det(A) = 1 \begin{vmatrix} 2 & 1 \\ 0 & 0 \end{vmatrix} - 3 \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + 0 \begin{vmatrix} 0 & 2 \\ 1 & 0 \end{vmatrix} = 1(0 - 0) - 3(0 - 1) + 0 = 3.$$

Step 2: Compute the cofactor matrix $C = [C_{ij}]$, where $C_{ij} = (-1)^{i+j} \det(M_{ij})$ and M_{ij} is the minor obtained by removing row i and column j .

$$C_{11} = \det \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} = 0, \quad C_{12} = -\det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 1, \quad C_{13} = \det \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} = -2,$$

$$C_{21} = -\det \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix} = 0, \quad C_{22} = \det \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = 0, \quad C_{23} = -\det \begin{bmatrix} 1 & 3 \\ 1 & 0 \end{bmatrix} = 3,$$

$$C_{31} = \det \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} = 3, \quad C_{32} = -\det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = -1, \quad C_{33} = \det \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix} = 2.$$

Step 3: Form the cofactor matrix

$$C = \begin{bmatrix} 0 & 1 & -2 \\ 0 & 0 & 3 \\ 3 & -1 & 2 \end{bmatrix}.$$

Step 4: Compute the adjugate matrix $\text{adj}(A) = C^T$

$$\text{adj}(A) = \begin{bmatrix} 0 & 0 & 3 \\ 1 & 0 & -1 \\ -2 & 3 & 2 \end{bmatrix}.$$

Step 5: Compute the inverse using

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = \frac{1}{3} \begin{bmatrix} 0 & 0 & 3 \\ 1 & 0 & -1 \\ -2 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{3} & 0 & -\frac{1}{3} \\ -\frac{2}{3} & 1 & \frac{2}{3} \end{bmatrix}.$$

(c) Since A is invertible, the system always has a unique solution:

$$x = A^{-1}b.$$

$$x = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{3} & 0 & -\frac{1}{3} \\ -\frac{2}{3} & 1 & \frac{2}{3} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ -\frac{2}{3} \\ \frac{10}{3} \end{bmatrix}.$$

Thus the system has a **unique solution**

$$x = \begin{bmatrix} 3 \\ -\frac{2}{3} \\ \frac{10}{3} \end{bmatrix}.$$

3. Let us consider the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 6 & 3 \\ 10 & 5 \\ 0 & 0 \end{bmatrix}.$$

- (a) (1 point) Compute the rank of A .
- (b) (1 point) Find a representation for the column space and a basis for $C(A)$.
- (c) (1 point) Find a representation for the row space and a basis for $C(A^T)$.
- (d) (1 point) What is the dimension of the left nullspace $N(A^T)$ and find a basis for it.
- (e) (1 point) What is the dimension of the nullspace $N(A)$ and find a basis for it.

Answer.

- (a) **Rank of A :** Step 1: Form the augmented matrix for row reduction (for column and row spaces):

$$A = \begin{bmatrix} 2 & 1 \\ 6 & 3 \\ 10 & 5 \\ 0 & 0 \end{bmatrix} \xrightarrow[r_3 \rightarrow r_3 - 5r_1]{r_2 \rightarrow r_2 - 3r_1} \begin{bmatrix} 2 & 1 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

$$\text{rank}(A) = 1.$$

- (b) **Column space $C(A)$:**

The pivot column corresponds to the first column of A , so

$$C(A) = \text{span} \left\{ \begin{bmatrix} 2 \\ 6 \\ 10 \\ 0 \end{bmatrix} \right\}.$$

- (c) **Row space $C(A^T)$:**

The nonzero row of the reduced row-echelon form corresponds to

$$\text{row vector: } \begin{bmatrix} 2 & 1 \end{bmatrix},$$

so

$$C(A^T) = \text{span} \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}.$$

- (d) **Left nullspace $N(A^T)$:**

Since $A \in \mathbb{R}^{4 \times 2}$ and $\text{rank}(A) = 1$,

$$\dim(N(A^T)) = m - \text{rank}(A) = 4 - 1 = 3.$$

To find a basis, solve $A^T y = 0$:

$$A^T = \begin{bmatrix} 2 & 6 & 10 & 0 \\ 1 & 3 & 5 & 0 \end{bmatrix}, \quad \begin{bmatrix} 2 & 6 & 10 & 0 \\ 1 & 3 & 5 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The equations are:

$$2y_1 + 6y_2 + 10y_3 = 0, \quad y_1 + 3y_2 + 5y_3 = 0.$$

The second equation is redundant. Let

$$y_2 = s, \quad y_3 = t, \quad y_4 = u.$$

Then $y_1 = -3s - 5t$, so a basis is

$$N(A^T) = \text{span} \left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

(e) **Nullspace** $N(A)$:

Since $\text{rank}(A) = 1$ and $n = 2$,

$$\dim(N(A)) = n - \text{rank}(A) = 1.$$

To find a basis, solve $Ax = 0$:

$$2x_1 + 1x_2 = 0 \quad \Rightarrow \quad x_1 = -\frac{1}{2}x_2.$$

Hence

$$N(A) = \text{span} \left\{ \begin{bmatrix} -\frac{1}{2} \\ 1 \end{bmatrix} \right\}.$$

4. Prove the following statements:

- (a) (1 point) If $A \in \mathbb{R}^{n \times n}$ and $\det(A) = a$, compute $\det((tA)^p + A^p)$ for $p \in \mathbb{N}$.
- (b) (1 point) If the matrix $A \in \mathbb{R}^{n \times n}$ has two equal rows, show that $\det(A) = 0$.
- (c) (3 point) Let $A \in \mathbb{R}^{m \times n}$ with $m \leq n$ and the system $Ax = b$ be solvable for every $b \in \mathbb{R}^m$.
 - Find $\dim(C(A))$, $\dim(C(A^T))$, $\dim(N(A))$, and $\dim(N(A^T))$.
 - Provide bases for $C(A^T)$ and $N(A)$.
 - How many solutions does the system $Ax = b$ have?

Answer.

- (a) Since $A \in \mathbb{R}^{n \times n}$ and $\det(A) = a$, we know that

$$\det((tA)^p + A^p) = \det(t^p \cdot A^p + A^p) = \det((t^p + 1)A^p) = (t^p + 1)^n \det(A^p) = (t^p + 1)^n (\det(A))^p = (t^p + 1)^n a^p.$$

- (b) If A has two equal rows, say row i and row j , then by the property of determinants that swapping two rows changes the sign:

$$\text{swap rows } i \text{ and } j \implies \det(A) = -\det(A).$$

Thus,

$$\det(A) = 0.$$

- (c) Since $Ax = b$ is solvable for every $b \in \mathbb{R}^m$, the columns of A span $\mathbb{R}^m$. Therefore, $C(A) = \mathbb{R}^m$. Thus,

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$$\dim(C(A)) = m, \quad \dim(C(A^T)) = m, \quad \text{rank}(A) = m,$$

$$\dim(N(A)) = n - \text{rank}(A) = n - m, \quad \dim(N(A^T)) = m - \text{rank}(A) = 0.$$

- Basis for $C(A^T)$: Take the rows of A . Basis for $N(A)$: Solve $Ax = 0$ to find $n - m$ independent vectors.
- Since $N(A)$ has dimension $n - m$, for each b the solution set is

$$x = x_0 + N(A).$$

If $m = n$, then there is a unique solution; and if $m < n$, there are infinitely many solutions.
